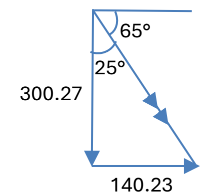
Questions			Answers	Marks
3	(a)		The torque of a force about an axis is the product of that force and the perpendicular distance from the line of action of the force to the axis.	B1
	(b)		Ty = Tcos20 = 410cos20 = 385.7 N Tx = Tsin20 = 410sin20 = 140.2 N Sum of forces in horizontal direction = 0, Rx = Tx = 140.23 N Sum of forces in vertical direction = 0, Ry = Ty – 65 – 20 = 300.27 N Resultant force at elbow = $(Rx^2 + Ry^2)^{0.5}$ = 331 N  $\tan^{-1}(140.23/300.27) = 25°$ Direction = 65° below the positive x-axis	M1 A1 C1 / A1
	(c)		Using Newton's 2nd law of motion, the large change in momentum and the short time duration of impact will result in greater force. Using Newton's 3rd law, the force on hand is equal in magnitude and opposite in direction to the force on wooden boards.	B1 B1